Week 1 Objective Proposal: Conceptual Design of a Lunar Regolith Transport Rover for Habitat Shielding Support

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1 Project Title

Conceptual Design of an Unmanned Lunar Regolith Transport Rover for Habitat Radiation Shielding Applications

2 Project Objective

This summer project aims to create a concept design for an unmanned lunar rover to transport lunar regolith to build shielding to protect a habitat from radiation. This project builds off previous research looking at using lunar regolith as a protective layer to lessen radiation exposure for astronauts. The objective of this is to understand the logistics of moving, transporting, and depositing regolith around a lunar habitat.

The design for the rover will be aimed at a system that is either fully or partially remote controlled or autonomous, and will possibly integrate LiDAR for navigation for terrain detection, obstacle avoidance, and positioning near the lunar habitat. The design will aim for a rover that can carry from 250 lb to 500 lb of regolith per trip and will consider 500 lb as a design goal. This size load is intended to make the rover implementable and usable for shielding the habitat and will ensure the rover design to be small and realistic for a conceptual system for use on the lunar surface.

3 Main Research Question

How can an unmanned lunar rover be designed to transport and deposit regolith efficiently for use in habitat radiation shielding systems?

4 Background and Motivation

Future lunar habitats will require protection from galactic cosmic rays, solar particle events, and other radiation hazards. Lunar regolith is a strong candidate for shielding because it is already available on the lunar surface, reducing the need to launch large quantities of shielding material from Earth. However, using regolith as shielding creates an important engineering challenge: the material must be excavated, transported, and placed around the habitat in a controlled and repeatable way.

This project focuses on the material transport stage of that problem. Instead of only studying how effective regolith is as a shielding material, this rover concept investigates how the regolith could physically be moved to the habitat. The proposed rover would carry regolith from a nearby collection zone to a shielding zone, where it could be dumped into trays, berms, or modular shielding structures around the habitat.

5 Connection to Previous Lunar Habitat Shielding Research

This project extends previous lunar habitat radiation shielding work by connecting the shielding design to a practical deployment system. In previous research, the protection layers of regoliths were analyzed as part of a lunar habitat protection strategy. This project asks how that regolith could be delivered to the habitat in the first place.

A possible use case is a rover that transports regolith to a multi-dome lunar habitat and deposits it into shielding trays or berms around the structure. In this way, the rover becomes part of a larger lunar base construction system. The project therefore combines mechanical design, lunar surface mobility, payload transport, and habitat integration.

6 Proposed Rover Mission Concept

The rover's main mission is to transport regolith from a collection area to a habitat shielding area. The rover does not need to fully excavate large amounts of regolith by itself during the first version of the project, but the design may include a front scoop, bucket drum, or assisted loading system as a future improvement.

6.1 Basic Mission Sequence

1. Rover begins at a charging or staging station near the lunar habitat.

2. Rover travels to a regolith collection or loading zone.

3. Regolith is loaded into the rover's storage bed or hopper.
4. Rover drives back to the habitat shielding area.
5. Rover deposits the regolith into a shielding tray, berm, or designated dump zone.
6. Rover returns to the collection zone and repeats the cycle.

7 Initial Payload Target

The initial design target is for the rover to carry up to 500 lb of regolith per trip. This value should be treated as a maximum target rather than a final optimized value. A more flexible engineering requirement is:

$$m_{payload} = 250\text{–}500 \text{ lb}$$

or, in SI units:

$$m_{payload} = 113\text{–}227 \text{ kg}$$

Since the rover operates on the Moon, the gravitational weight of this payload is lower than on Earth. The lunar gravitational acceleration is approximately:

$$g_{moon} = 1.62\,\mathrm{m/s^2}$$

The lunar weight of a 227 kg payload is:

$$W_{moon} = mg_{moon}$$

$$W_{moon} = (227)(1.62) = 367.7\,\mathrm{N}$$

Converting this to pounds-force:

$$W_{moon} \approx 82.7\,\mathrm{lbf}$$

Therefore, a 500 lb Earth-mass payload would only weigh about 83 lbf on the Moon. However, the rover still has to handle the full mass and inertia of the payload, so the structure, motors, wheels, and braking system must still be designed carefully.

8 Rover Configuration

8.1 General Layout

The proposed rover will use a low, rectangular chassis with a central dump bed or hopper. The low center of gravity will improve stability while carrying regolith. The rover will use multiple wheels on each side to increase surface contact, improve traction, and reduce sinkage in soft lunar soil.

A possible initial configuration is:

- Six-wheel or eight-wheel rover layout
- Three or four wheels on each side
- Independent wheel motors for improved torque control
- Rocker-bogie or articulated suspension concept
- Central dump bed for regolith storage
- Rear or side dumping mechanism
- Solar charging system as the main power source
- Battery storage for operation when solar input is reduced
- Optional LiDAR sensor mast for navigation and terrain mapping

9 Wheel Design Ideas

Wheel design is one of the most important parts of the rover because lunar regolith is loose, dusty, and deformable. The rover should avoid narrow smooth wheels because they may sink into the soil and lose traction. Instead, the rover should use wide wheels with high contact area and strong surface features.

9.1 Recommended Wheel Features

- **Wide wheel profile:** Increases contact area and reduces sinkage.
- **Metallic or flexible mesh structure:** Reduces mass while allowing the wheel to deform slightly over rough terrain.
- **Grousers or cleats:** Raised ridges on the wheel surface help the rover grip loose regolith.

- **Independent wheel drive:** Each wheel can receive torque separately, improving mobility if one wheel slips.
- **Large wheel diameter:** Helps the rover climb over rocks and small terrain obstacles.
- **Dust-tolerant bearings and seals:** Reduces damage from abrasive lunar dust.

9.2 Proposed Wheel Concept

The preferred wheel concept is a wide metallic grouser wheel. Each wheel would include raised transverse ribs that dig slightly into the regolith as the wheel rotates. This increases traction by creating mechanical interaction between the wheel and the soil. The wheels should be large enough to reduce rolling resistance and to allow the rover to climb small rocks or uneven terrain near the habitat.

A six-wheel or eight-wheel system is preferred over a four-wheel system because the rover is expected to carry a heavy regolith payload. More wheels distribute the load over a larger contact area, which reduces sinkage and improves stability.

10 Suspension and Structural Support

The rover should include braces and suspension components to improve stability and protect the chassis during motion over uneven terrain. Since the rover may be carrying up to 500 lb of regolith mass, the frame must resist bending and twisting.

10.1 Suspension Concept

A rocker-bogie inspired suspension or independent trailing-arm suspension may be used. The purpose of the suspension is not passenger comfort, but terrain adaptability. The wheels should remain in contact with the ground even when the rover travels over rocks, slopes, or small craters.

10.2 Structural Bracing

The rover chassis should include diagonal braces to improve stiffness. These braces would connect the lower chassis to the upper support frame around the dump bed. The braces would help distribute payload forces into the wheel assemblies and reduce bending stress in the central frame.

Important structural features include:

- Rectangular aluminum or titanium frame concept

- Cross-bracing under the dump bed
- Side braces connecting suspension mounts to the main chassis
- Reinforced hinge region for the dump bed
- Low center of gravity to prevent tipping

11 Regolith Collection and Dumping Mechanism

The rover will mechanically function similar to a small lunar dump truck. Its main role is to carry regolith in a storage bed and deposit it near or onto the habitat shielding system.

11.1 Loading Options

There are three possible loading concepts:

1. **External loading:** A separate excavator loads regolith into the rover.
2. **Front scoop:** The rover uses a front scoop or blade to collect regolith.
3. **Bucket drum system:** A rotating drum with small scoops collects regolith and transfers it into the hopper.

For the first version of this project, external loading or a front scoop is the simplest and most realistic option. A bucket drum system is more advanced and could be included as a future design improvement.

11.2 Dumping System

The rover will use a tilting dump bed to release regolith. The bed would rotate about a rear hinge, similar to a dump truck. A linear actuator or screw-driven lift mechanism would raise the front of the bed, allowing regolith to slide out of the rear opening.

The dumping sequence would be:

1. Rover positions itself near the habitat shielding tray or berm.
2. Rear gate opens or remains passively hinged.
3. Linear actuator raises the front of the dump bed.
4. Regolith slides out due to gravity.
5. Dump bed lowers back into the locked transport position.

6. Rover returns to the collection area for another load.

The dump angle should be large enough for dry regolith to slide out. A preliminary target dump angle is:

$$\theta_{dump} = 35^\circ\text{–}50^\circ$$

This angle may be refined later based on assumptions about regolith friction and bed material.

12 Power System

The rover will use solar charging as its main power source. Solar panels may be mounted on the top of the rover, above the chassis, or on a deployable panel system. Since solar power is not always constant, the rover should also include a rechargeable battery system.

12.1 Power System Components

- Solar panels for primary energy generation
- Battery pack for energy storage
- Electric motors for wheel drive
- Electric actuator for dump bed lifting
- Power management electronics
- Sensors such as cameras, LiDAR, and inertial measurement units

13 Remote Control and LiDAR-Based Navigation

The rover may be designed as either remotely controlled or semi-autonomous. A remote-controlled version is simpler and would allow an operator to command the rover from a habitat or control station. A LiDAR-based version would allow the rover to detect obstacles, estimate terrain shape, and navigate between the collection zone and the habitat.

13.1 Possible Sensor Suite

- LiDAR for terrain mapping and obstacle detection
- Stereo cameras for visual navigation

- IMU for orientation and motion tracking
- Wheel encoders for distance estimation
- Load sensors to estimate regolith payload

14 Initial Engineering Requirements

Requirement Category	Initial Requirement
Environment	Rover must be designed for conceptual lunar surface operation.
Primary Function	Transport regolith to habitat shielding locations.
Payload Capacity	Carry approximately 250–500 lb of regolith per trip.
Mobility	Use six or eight wheels with high-traction wheel design.
Suspension	Include suspension or articulated wheel supports for uneven terrain.
Structure	Include bracing to support payload and reduce chassis bending.
Power	Use solar charging with battery storage.
Navigation	Support remote operation with possible LiDAR-based autonomy.
Unloading	Use a dump-bed mechanism similar to a small dump truck.
Habitat Integration	Deposit regolith into shielding trays, berms, or habitat protection zones.

Table 1: Initial rover design requirements.

15 Key Calculations for Later Project Phases

15.1 Regolith Payload Mass

$$m_{regolith} = \rho_{regolith} V_{hopper}$$

where $m_{regolith}$ is the mass of regolith, $\rho_{regolith}$ is the bulk density of regolith, and V_{hopper} is the volume of the rover hopper.

15.2 Number of Trips Required

$$N = \frac{M_{required}}{m_{payload}}$$

where N is the number of trips, $M_{required}$ is the total regolith mass needed for shielding, and $m_{payload}$ is the mass carried per rover trip.

15.3 Lunar Weight of Payload

$$W_{moon} = m_{payload} g_{moon}$$

where $g_{moon} = 1.62\,\mathrm{m/s^2}$.

15.4 Wheel Traction Force

$$F_{traction} = \mu N$$

where μ is the wheel-regolith traction coefficient and N is the normal force on the wheel.

15.5 Power Required for Motion

$$P = Fv$$

where P is power, F is the required driving force, and v is rover velocity.

15.6 Energy Per Trip

$$E = Pt$$

where E is energy, P is power, and t is trip time.

16 Conclusion

This project proposes the conceptual design of an unmanned lunar regolith transport rover that supports habitat radiation shielding. The rover would carry approximately 250–500 lb of regolith per trip, use multiple high-traction wheels, include suspension and structural bracing, rely on solar charging with battery storage, and deposit regolith using a dump-bed mechanism. The project connects directly to previous lunar habitat shielding work by addressing the practical question of how regolith can be moved and placed around a habitat. By the end of the summer, the project will produce a complete engineering design study including CAD, calculations, system requirements, and a final report.

References

[1] NASA Kennedy Space Center, "NASA Kennedy Digs Latest Robot Test," 2025. RASSOR regolith excavation and dumping concept.

[2] NASA Science, "VIPER Rover and Instruments." Reference for lunar rover wheel modules, active suspension, and mobility concepts.

[3] NASA, "Help NASA Design a Robot to Dig on the Moon," 2020. Reference for lunar regolith excavation systems and ISRU motivation.